LAB 04 | Recorded FPS Results

Interactive Fragment Shaders | AMD Radeon vs NVIDIA RTX 4050

Source: two user-recorded screen captures of the Stanford Bunny OpenGL application, captured on the same ASUS gaming laptop. The readings below were transcribed from the application window title at specified timestamps.

Scope: 10, 100, 1,000 and 10,000 Bunnies; Solid Color, Checkerboard and Fireflies shader modes. Both recordings show interactive controls and animated/rotating Bunnies. The screen recorder runs at 30 FPS, which is unrelated to the OpenGL FPS counter.

How to interpret the measurements

Every value is a **single visible on-screen FPS reading** at the indicated second, not a timed mean, minimum, or GPU-only rendering time. Modes and visual parameters are changed during the recording, so cross-GPU comparisons are illustrative, not a controlled benchmark. Some effects may have different colors or settings between captures.

10 Bunnies — 49,680 triangles per frame

Shader mode	AMD FPS (time)	RTX 4050 FPS (time)	RTX / AMD
Solid Color	773.2 (2s)	826.1 (14s)	1.07x
Checkerboard	870.2 (12s)	690.7 (2s)	0.79x
Fireflies	Not verified	737.4 (22s)	—

AMD Fireflies at 10 Bunnies: no unambiguous, steady frame was selected from the sampled footage; left unreported rather than guessed.

100 Bunnies — 496,800 triangles per frame

Shader mode	AMD FPS (time)	RTX 4050 FPS (time)	RTX / AMD
Solid Color	627.7 (28s)	871.6 (38s)	1.39x
Checkerboard	610.8 (32s)	859.4 (32s)	1.41x
Fireflies	674.4 (24s)	914.2 (44s)	1.36x

1,000 Bunnies — 4,968,000 triangles per frame

Shader mode	AMD FPS (time)	RTX 4050 FPS (time)	RTX / AMD
Solid Color	204.2 (44s)	561.1 (56s)	2.75x
Checkerboard	199.9 (40s)	532.1 (50s)	2.66x
Fireflies	203.1 (48s)	517.4 (60s)	2.55x

10,000 Bunnies — 49,680,000 triangles per frame

Shader mode	AMD FPS (time)	RTX 4050 FPS (time)	RTX / AMD
Solid Color	44.6 (64s)	67.2 (78s)	1.51x

Shader mode	AMD FPS (time)	RTX 4050 FPS (time)	RTX / AMD
Checkerboard	45.3 (68s)	67.1 (68s)	1.48x
Fireflies	45.1 (58s)	67.1 (84s)	1.49x

Observations

At 1,000 Bunnies, the sampled RTX readings are about 2.5–2.8 times the AMD readings across the three shader modes. At 10,000 Bunnies, the selected RTX readings are about 1.5 times the AMD readings. These are observations from screen recordings, not definitive GPU performance ratios.

Recording and extraction notes

AMD recording: 69.0 seconds, 1906 × 1198 at 30 capture FPS. RTX recording: 87.5 seconds, 1916 × 1198 at 30 capture FPS. The footage shows application title "Lab03" while its interactive control panel reads "Lab 04". A brief intermediate count (433) appears in AMD footage; it is excluded. The number 4,968 triangles per Bunny is shown in the panel.

Limitations and recommended next experiment

A fair benchmark should fix the resolution, shader parameters, rotation, power mode, VSync, and build type, run without screen recording, and collect repeated timed averages plus 1% low FPS and CPU/GPU timing. The present table deliberately preserves the visible recorded values without inventing missing data.

Visual evidence from the recordings

Selected source frames showing the FPS title bar, shader controls, and Bunny grid.

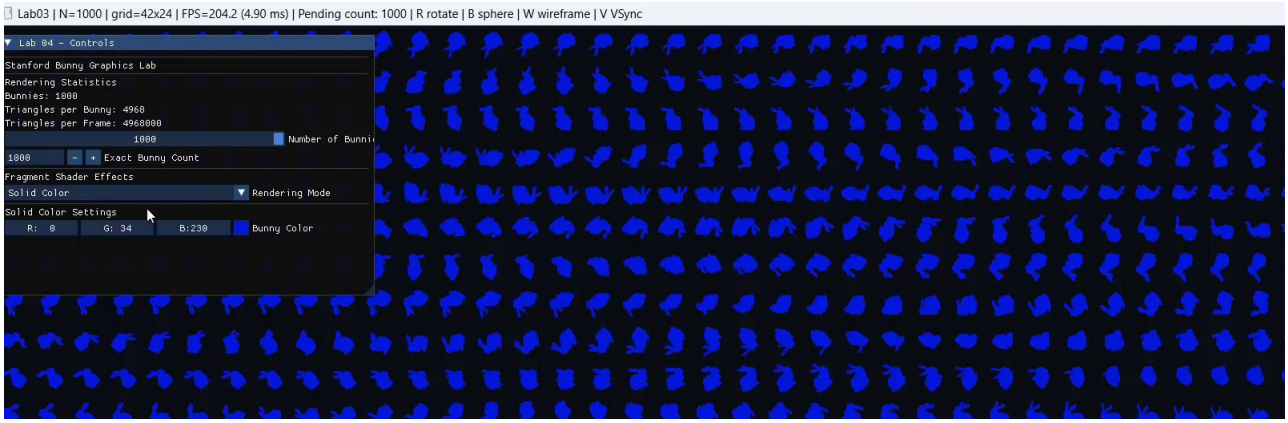
AMD Radeon — 10 Bunnies / Checkerboard / 870.2 FPS



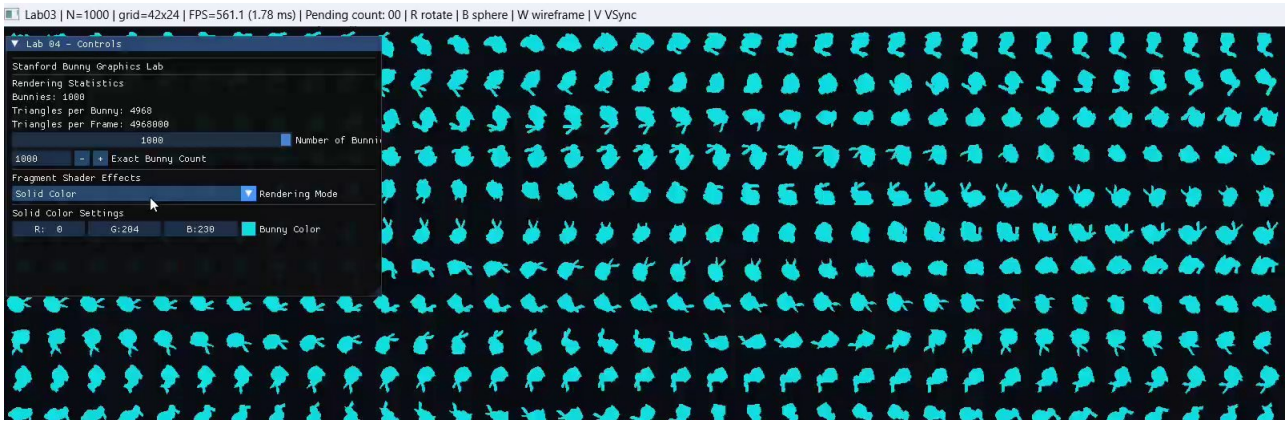
RTX 4050 — 10 Bunnies / Solid Color / 826.1 FPS



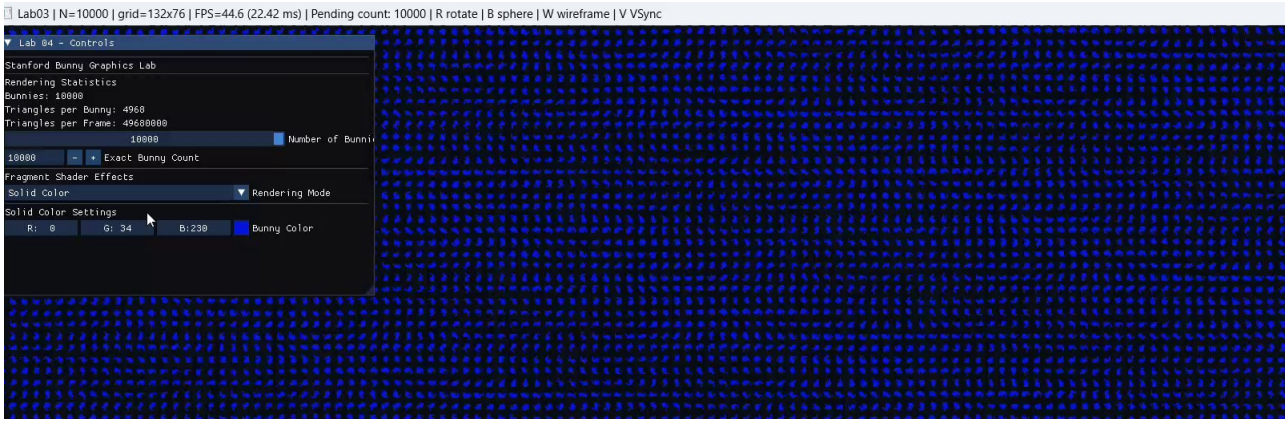
AMD Radeon — 1,000 Bunnies / Solid Color / 204.2 FPS



RTX 4050 — 1,000 Bunnies / Solid Color / 561.1 FPS



AMD Radeon — 10,000 Bunnies / Solid Color / 44.6 FPS



RTX 4050 — 10,000 Bunnies / Solid Color / 67.2 FPS

